Assignment 4

Due 11:59 p.m., February 22, 2026

> **IMPORTANT!** This is an individual assignment. You may discuss broad issues of interpretation, understanding, and general approaches to a solution. However, the development of a specific solution or program code must be your own work. The assignment is expected to be entirely your own, designed and coded by you alone. If you need assistance, please consult your instructor or the TAs. Be sure to read the specific policies outlined in the **Academic Honesty in Computing** section.

Problem Definition

Using UDP-based sockets, write a C program, `myping.c`, which mimics the standard `ping` command as follows:

```
$ ./myping syslab01.cs.trincoll.edu
from syslab01.cs.trincoll.edu (157.252.16.50): time=0.620 ms
...
```

where `time` is the Round-Trip Time (RTT), the delay between the client sending the ping message and receiving the pong message. The output must mirror the standard Unix `ping` format, requiring precision in millisecond calculations.

To test your program, write another C program, `myping_server.c`, which runs on the server to respond to the client's ping request. The server must be a persistent process that echoes requests back to the source. It must bind to a specific port number (e.g., a non-well-known port above 1024) to listen for incoming packets. The server should capture the client's socket address so it knows exactly where to send the pong packet.

Programming Notes

1. Use a Datagram Socket (`SOCK_DGRAM`) to send packets.
2. Because UDP is unreliable, you must handle the possibility of a lost packet. Hint: Use `setsockopt()` with the `SO_RCVTIMEO` flag.
3. Use `gethostbyname()` to convert the hostname (e.g., `syslab01`) into the 32-bit IP address required for the socket address.
4. Use `gettimeofday()` immediately before `sendto()` and immediately after `recvfrom()` to calculate the RTT.

Handin

When completed, submit your C source, `myping.c` and `myping_server.c`, on the course website.

- **Welcome: Sean**
  - LogOut

Trinity College
HARTFORD CONNECTICUT